MILITARY SPECIFICATION SHEET

INTERCOMMUNICATIONS SETS, AN/AIC-14 and AN/AIC-14A

AUDIO FREQUENCY AMPLIFIER

AM-3889/AIC-14A

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope. This specification sheet provides performance and physical characteristics for the Audio Frequency Amplifier AM-3889/AIC-14A, hereinafter referred to as AM-3889.

1.2 Military Part Number. The military part number shall be M22352/24 -1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

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3. REQUIREMENTS

3.1 Item Definition. The Audio Frequency Amplifier AM-3889 is normally used in conjunction with the C-2645D master Interphone Control panel. The AM-3889 shall be capable of accepting two separate microphone inputs. The AM-3889 shall provide amplified voice and sidetone output signals to headsets at two different crew stations. A two-channel output shall be provided to permit the use of two headsets without interaction or interference. The outputs shall be capable of operating directly into high impedance headsets, low impedance headsets, or loudspeakers without the use of external impedance matching transformers.

3.1.1 Item Diagram. The Audio Frequency Amplifier shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The Audio Frequency Amplifier input and output signals will normally display the characteristics presented in Table I.

TABLE I. Interface signals.

Signal Name	Characteristics
Inputs:	
Microphone No. 1 (MIC NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Microphone No. 2 (MIC NO. 2)	Same as Mic No. 1
Audio (AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Power (PWR)	+27.5 Vdc ±0.5V
Outputs:	
Speaker No. 1 (SPKR NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 3.87 Vac. Signal power range is 0 to 2 Watts.
Speaker No. 2 (SPKR NO. 2)	Same as Speaker No. 1

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TABLE I. Interface signals - continued

Signal Name	Characteristics
Headset (HDST)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 13.4 Vac.
Mic Audio (MIC AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging from 0 to 1.5 Vac.
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	System return. See 3.4.4.3 of MIL-I-22352(AS).
Mic Return (MIC RTN)	Signal return to system ground.
Headset Return (HDST RTN)	Signal return to system ground.

3.1.3 SRAs. The AM-3889 shall contain the SRAs listed in Table II.

Title	Detail Spec.
Microphone Mixing Module	MIL-I-22352/4(AS).
Dual Output Amplifier	MIL-I-22352/8(AS).

A brief description of each SRA's function is given below.

3.1.3.1 Microphone Mixing Module. The microphone mixing module provides amplification to two independent audio signals and combines these two signals to provide a single output. The output shall be capable of supplying at least 14 milliwatts into a 100 ohm load. The module also contains a low pass filter for attenuation of ripple on the microphone current source. Performance shall be in accordance with MIL-I-22352/4(AS).

3.1.3.2 Dual Output Amplifier. The dual output amplifier provides two power amplifiers whose inputs are driven in parallel and whose outputs are capable of being independently switched to separate output loads. The outputs shall each be capable of supplying at least 2 watts into a 7.5 ohm load. Performance shall be in accordance with MIL-I-22352/8(AS).

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3.1.4 Controls. The audio frequency amplifier shall provide the controls listed in Table III.

TABLE III. AM-3889 controls list.

Title	Function
SPKR-HDST	The AM-3889 shall provide two rotary three position switches each with the following functions: SPKR 7.5 - In the CCW position the AM-3889 audio output shall be directed to the associated 7.5 ohm speaker with signal power and impedance matching characteristics as defined herein. HDST 7.5 - In the center position the AM-3889 audio output shall be directed to the associated headset with signal power and impedance matching characteristics as defined herein. HDST 600 - In the CW position the AM-3889 audio output shall be directed to the associated 600 ohm headset with signal power and impedance matching characteristics as defined herein.

3.2 Characteristics

3.2.1 Performance characteristics.

3.2.1.1 Input characteristics. The audio frequency amplifier shall have the following input characteristics:

3.2.1.1.1 Microphone No. 1 (MIC NO. 1). The MIC NO. 1 input shall accept signals generated by a microphone or microphone circuit. The input shall have the following characteristics:

3.2.1.1.1.1 Signal Frequency. The AM-3889 shall accept MIC NO. 1 input drive signals which range from 300 to 7000 Hz.

3.2.1.1.1.2 Signal Amplitude. The AM-3889 shall accept MIC NO. 1 input drive signals which range in amplitude from 0.01 to 0.10 Vac. The AM-3889 shall be capable of withstanding overvoltage inputs of up to 1.00 Vac without sustaining damage.

3.2.1.1.1.3 Input Impedance. The impedance of each MIC NO. 1 input shall be 1200 ohms ± 10 percent.

3.2.1.1.1.4 Input Isolation. The isolation between the MIC NO. 1 and the MIC NO. 2 inputs shall not be less than 20 dB.

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3.2.1.1.2 Microphone No. 2 (MIC NO. 2). The MIC NO. 2 input shall have characteristics equivalent to those of MIC NO. 1 as described in Section 3.2.1.1.1.

3.2.1.1.3 Audio (AUDIO). The AUDIO input shall accept signals from the interphone lines and have the following characteristics:

3.2.1.1.3.1 Signal Frequency. The AM-3889 shall accept AUDIO input drive signals which range from 300 to 7000 Hz.

3.2.1.1.3.2 Signal Amplitude. The AM-3889 shall accept AUDIO input drive signals which range in amplitude from 0.01 to 0.10 Vac. The AM-3889 shall be capable of withstanding overvoltage inputs of up to 1.00 Vac without sustaining damage.

3.2.1.1.3.3 Input Impedance. The impedance of the AUDIO input shall be 1200 ohms ± 10 percent.

3.2.1.1.4 Power (PWR). The Audio Frequency Amplifier shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the Audio Frequency Amplifier shall not exceed 500 milliamperes.

3.2.1.2 Output characteristics. All signals which are passed through this unit from input to output by being routed only through connectors, wire, printed wiring, switches, and relays shall not experience insertion loss in excess of 0.5 dB. The Audio Frequency Amplifier shall have the following output characteristics:

3.2.1.2.1 Speaker No. 1 (SPKR NO. 1). The SPKR NO. 1 output shall have characteristics which comply with the following description:

3.2.1.2.1.1 Signal Frequency. The output shall not exceed ±1dB from the 1,000 Hz output value over the range of frequencies from 300 Hz to 7,000 Hz. Further the output shall not exceed the 1000 Hz reference level for any frequency above 10000 Hz.

3.2.1.2.1.2 Signal Amplitude/Gain. The output Power shall be 8 ±2 dB above the input for speaker operation and 0±2 dB above the input for headset operation. The input/output characteristic (gain) shall be linear within ±1 dB for input levels between 3.0 and 12.0 volts and linear within ±2 dB for input levels between 1.2 and 3.0 volts.

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3.2.1.2.1.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall not exceed the limits established in Table IV.

TABLE IV. Distortion limits.

Environmental Condition	Output Level (watts)	Distortion Limit (percent)
Standard	1	5
All environmental	1	10
Standard	2	10
All environmental	2	15

Where:
Standard - Standard conditions defined in MIL-T-5422.
All - Environmental envelope defined in MIL-E-5400 for Class II equipment.

3.2.1.2.1.4 Output Power. The output of the amplifier shall be capable of delivering 2 watts into a 7.5 ohm resistive load.

3.2.1.2.1.5 Amplifier Stability. The amplifier shall operate with no evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short, or complex impedance terminations. The amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.2.1.2.1.6 Internal Noise. The amplifier shall have an output of less than 5 millivolts throughout the environmental envelope in MIL-E-5400 for Class II equipment when operated with the input shorted.

3.2.1.2.1.7 Output Impedance. The amplifier shall be capable of delivering the specified output power when terminated in a 7.5 ohm resistive load.

3.2.1.2.2 Speaker No. 2 (SPKR NO. 2). The SPKR NO. 2 output shall have characteristics equivalent to those of SPKR NO. 1 as described in Section 3.2.1.2.1.

3.2.1.2.3 Headset No. 1 (HDST NO. 1). There shall be two HDST HIGH NO. 1 outputs available in the AM-3889. Each HDST HIGH output shall have characteristics which comply with the following description.

3.2.1.2.3.1 Signal Frequency. The output shall not exceed ±1 dB from the 1,000 Hz output value over the range of frequencies from 300 Hz to 7,000 Hz. Further the output shall not exceed the 1,000 Hz reference level for any frequency above 10,000 Hz.

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3.2.1.2.3.2 Signal Amplitude/Gain. The output power shall be 0 ± 2 dB above the input. The input/output characteristic (gain) shall be linear within ±1 dB for input levels between 3.0 and 12.0 volts rms and linear within ±3 dB for input levels between 1.2 and 3 volts rms.

3.2.1.2.3.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall not exceed 5 percent for standard conditions as defined by section 3.1a of MIL-STD-810 at 300 milliwatts output and 8 percent through the environmental envelope specified herein.

3.2.1.2.3.4 Output Power. The output of the amplifier shall be capable of delivering 300 milliwatts into a 600 ohm resistive load when the HDST 600 position has been selected on the SPKR-HDST control or when HDST 7.5 has been selected on the SPKR-HDST control and the output terminated in a 7.5 ohm non-inductive load.

3.2.1.2.3.5 Amplifier Stability. The amplifier shall operate with no evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short, or complex impedance terminations. The amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.2.1.2.3.6 Internal Noise. The amplifier shall have an output of less than 5 millivolts throughout the environmental envelope defined in MIL-E-5400 for Class II equipment when operated with the input shorted.

3.2.1.2.3.7 Output Impedance. The amplifier shall be capable of delivering the specified output power when terminated in a 7.5 ohm resistive load.

3.2.1.2.4 Headset No. 2 (HDST NO. 2). The performance characteristics displayed by the two HDST NO. 2 outputs shall be the same as those described in Section 3.2.1.2.3 for HDST NO. 1.

3.3 PHYSICAL CHARACTERISTICS

3.3.1 Dimensions. The size of the unit is shown in Figure 2. All output dimensions shall be within ±0.03 inches of the nominal value shown.

3.3.2 Weight. The AM-3889 shall weight no more than 2 pounds.

3.4 DESIGN AND CONSTRUCTION

The design and construction of the Audio Frequency Amplifier shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 of MIL-I-22352(AS).

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4.3 Test Methods.

4.3.1 Examination of product. See 4.3.3.2.1 of MIL-I-22352(AS).

4.3.2 Operational test. See 4.3.3.2.2 of MIL-I-22352(AS).

4.3.2 Input Impedance. Verify that the input impedances are in accordance with 3.2.1.1.1.3 and 3.2.1.1.3.3 herein.

4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.2.1.2.1.2 and 3.2.1.2.3.2 herein.

4.3.2.3 Frequency Response. Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements of 3.2.1.2.1.1 herein.

4.3.2.4 Linearity. Using a 1000 Hz input signal which is varied, verify that the amplifier meets the requirements of 3.2.1.2.1.2.

4.3.2.5 Output Power. With the amplifier properly terminated, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.2.1.2.1.4 and 3.2.1.2.3.4 herein.

4.3.2.6 Distortion. Verify the requirements of 3.2.1.2.1.3 herein.

4.3.2.7 Amplifier stability. Verify the requirements of 3.2.1.2.1.5 herein.

4.3.2.8 Internal noise. Verify the requirements of 3.2.1.2.1.6 herein.

4.3.2.9 Susceptibility to stray magnetic fields. Verify the performance of 3.2.1.2.1.8 herein.

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3.2.1.2.3.2 Signal Amplitude/Gain. The output power shall be 0 ± 2 dB above the input. The input/output characteristic (gain) shall be linear within ±1 dB for input levels between 3.0 and 12.0 volts rms and linear within ±3 dB for input levels between 1.2 and 3 volts rms.

3.2.1.2.3.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall not exceed 5 percent for standard conditions as defined by section 3.1a of MIL-STD-810 at 300 milliwatts output and 8 percent through the environmental envelope specified herein.

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3.2.1.2.3.5 Amplifier Stability. The amplifier shall operate with no evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short, or complex impedance terminations. The amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

3.2.1.2.3.6 Internal Noise. The amplifier shall have an output of less than 5 millivolts throughout the environmental envelope defined in MIL-E-5400 for Class II equipment when operated with the input shorted.

3.2.1.2.3.7 Output Impedance. The amplifier shall be capable of delivering the specified output power when terminated in a 7.5 ohm resistive load.

3.2.1.2.4 Headset No. 2 (HDST NO. 2). The performance characteristics displayed by the two HDST NO. 2 outputs shall be the same as those described in Section 3.2.1.2.3 for HDST NO. 1.

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4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.2.1.2.1.2 and 3.2.1.2.3.2 herein.

4.3.2.3 Frequency Response. Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements of 3.2.1.2.1.1 herein.

4.3.2.4 Linearity. Using a 1000 Hz input signal which is varied, verify that the amplifier meets the requirements of 3.2.1.2.1.2.

4.3.2.5 Output Power. With the amplifier properly terminated, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.2.1.2.1.4 and 3.2.1.2.3.4 herein.

4.3.2.6 Distortion. Verify the requirements of 3.2.1.2.1.3 herein.

4.3.2.7 Amplifier stability. Verify the requirements of 3.2.1.2.1.5 herein.

4.3.2.8 Internal noise. Verify the requirements of 3.2.1.2.1.6 herein.

4.3.2.9 Susceptibility to stray magnetic fields. Verify the performance of 3.2.1.2.1.8 herein.

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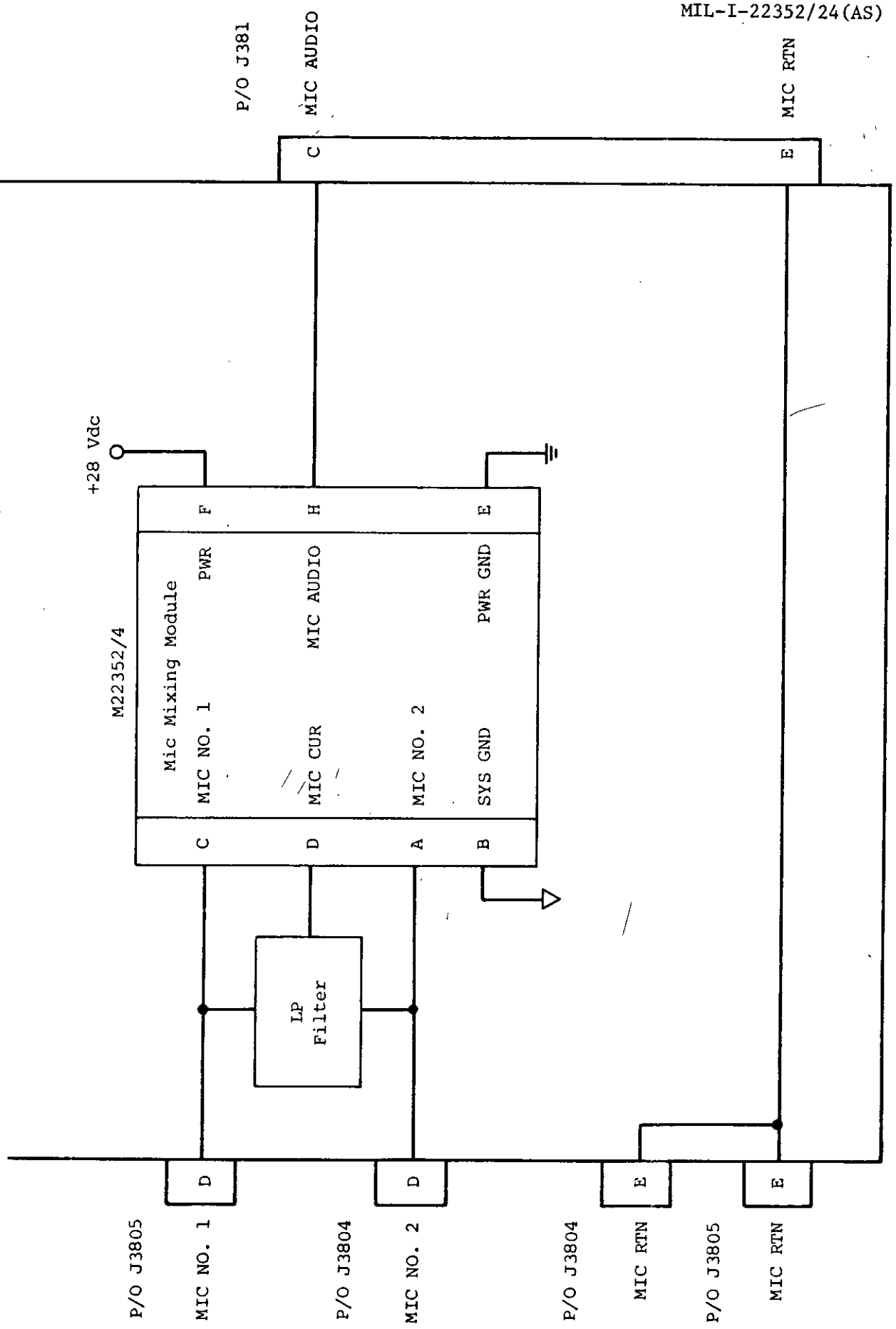
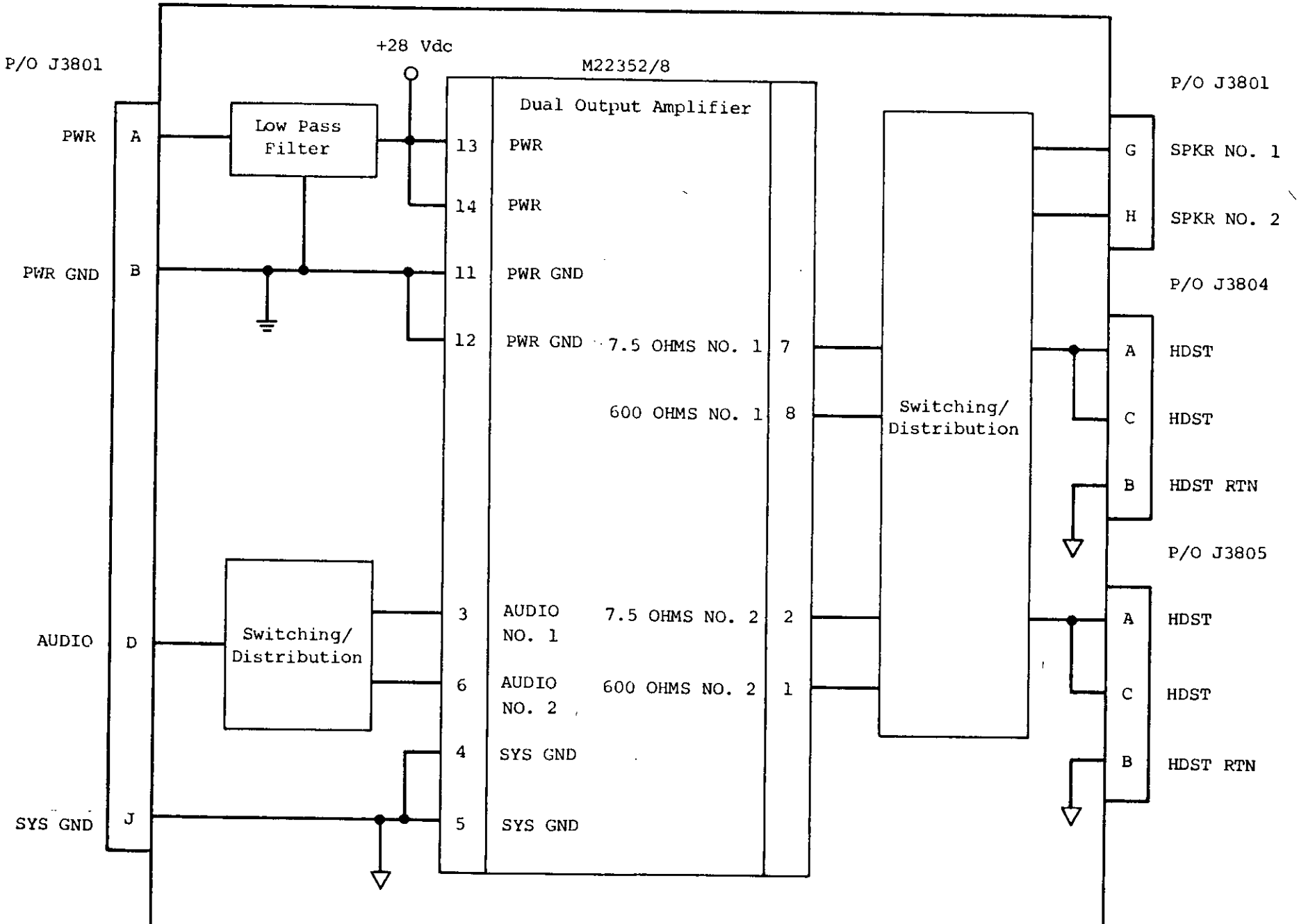
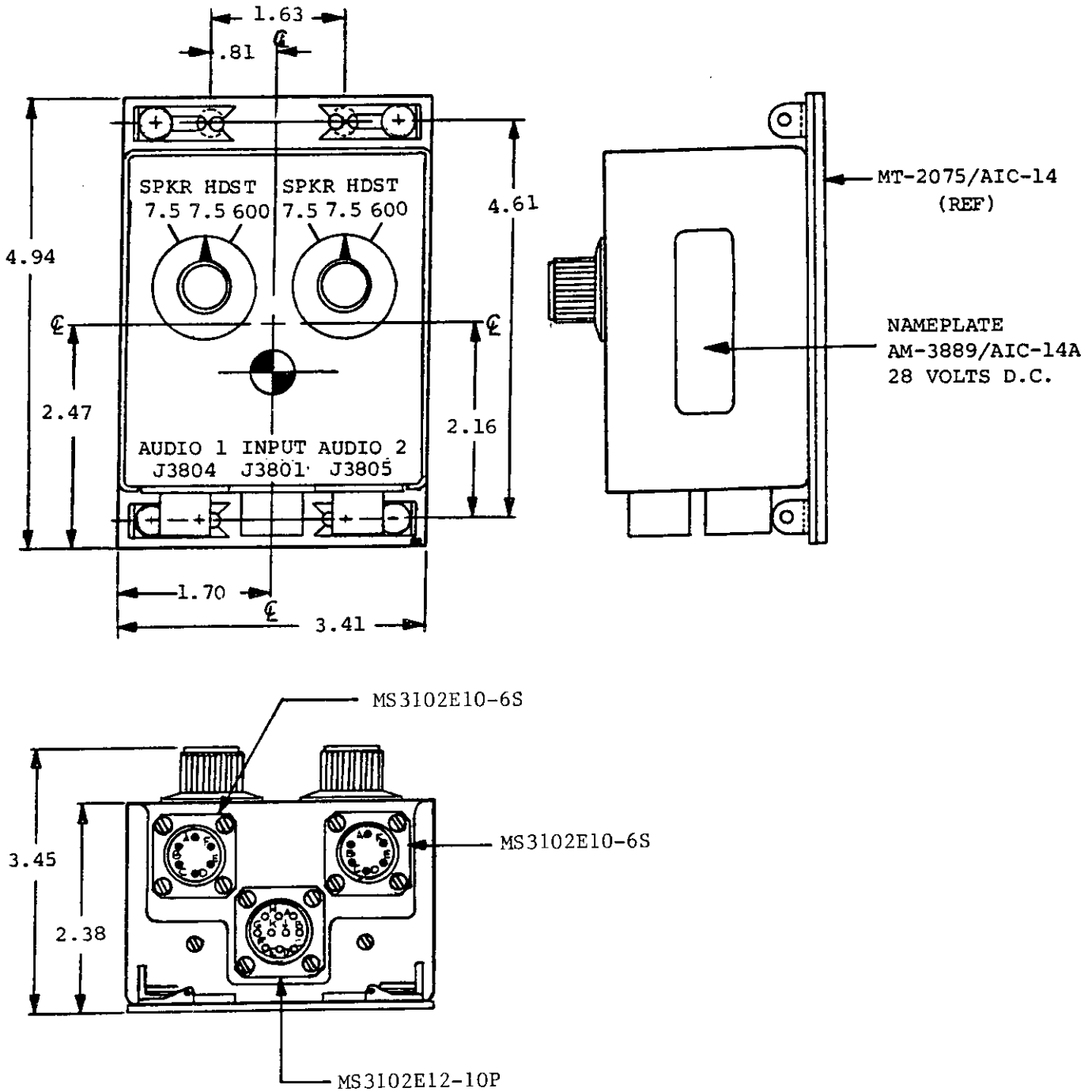


FIGURE 1. AM-3889 functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. AM-3889 outline drawing.

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